

China Mobile Use Case: Network Changes enabled by Agent and Network Digital Twin

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Challenges in Traditional Network Change Scenario

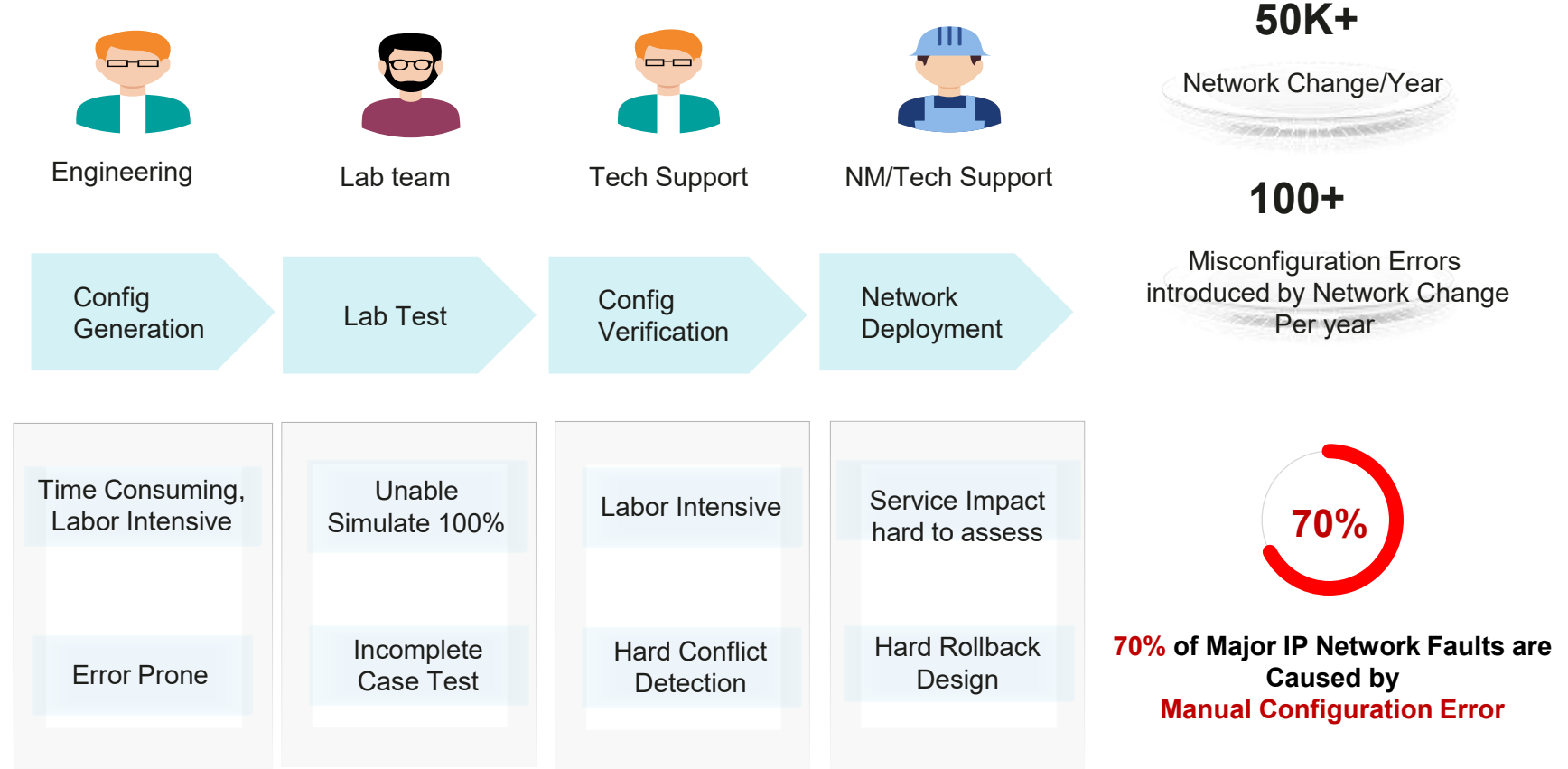
Native Characteristics of IP Networks

Large in scale

Single failure can affect whole network

High risk of service interruption

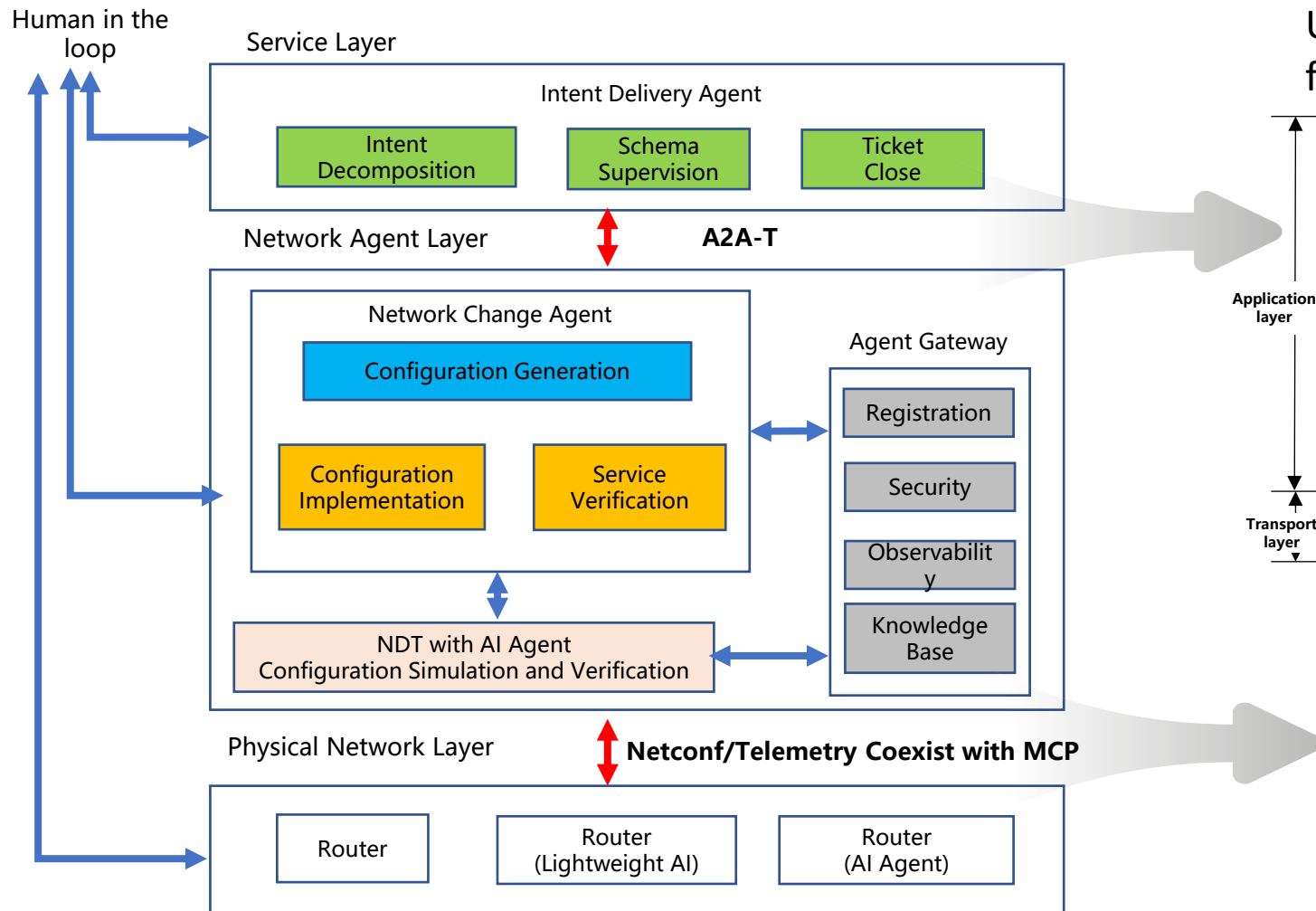
Traditional Network Change Scenario



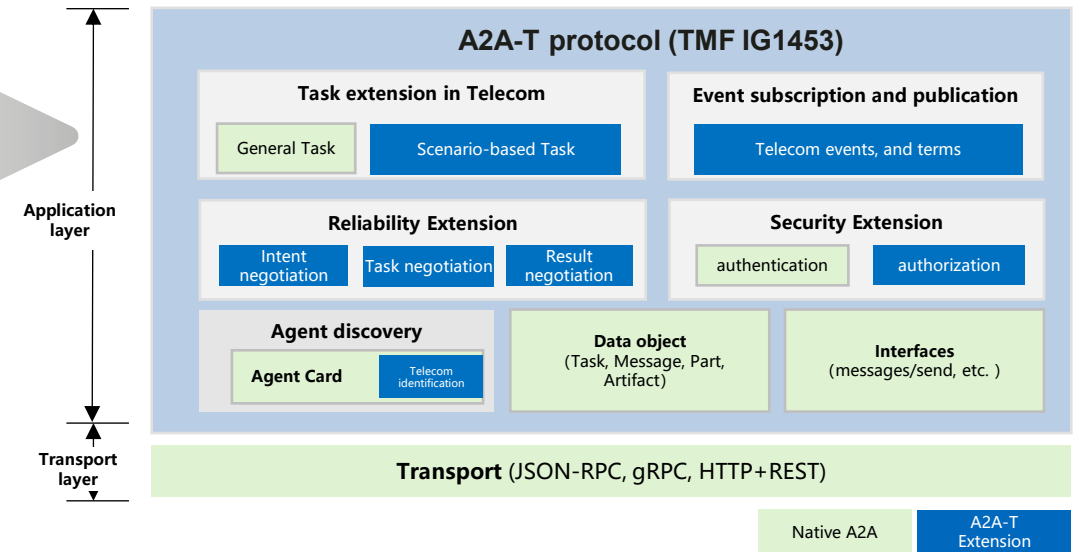
- Challenge 1: Manual configuration is low-efficiency, cannot meet the service need.
- Challenge 2: Rely on manual experience, may lead to network faults, cannot guarantee service security.

CM Agentic Architecture: combined with agent and NDT

- AI Agent: intent understanding and network configuration generation.
- NDT: configuration simulation. Using agent to reduce modeling overhead and improve accuracy.

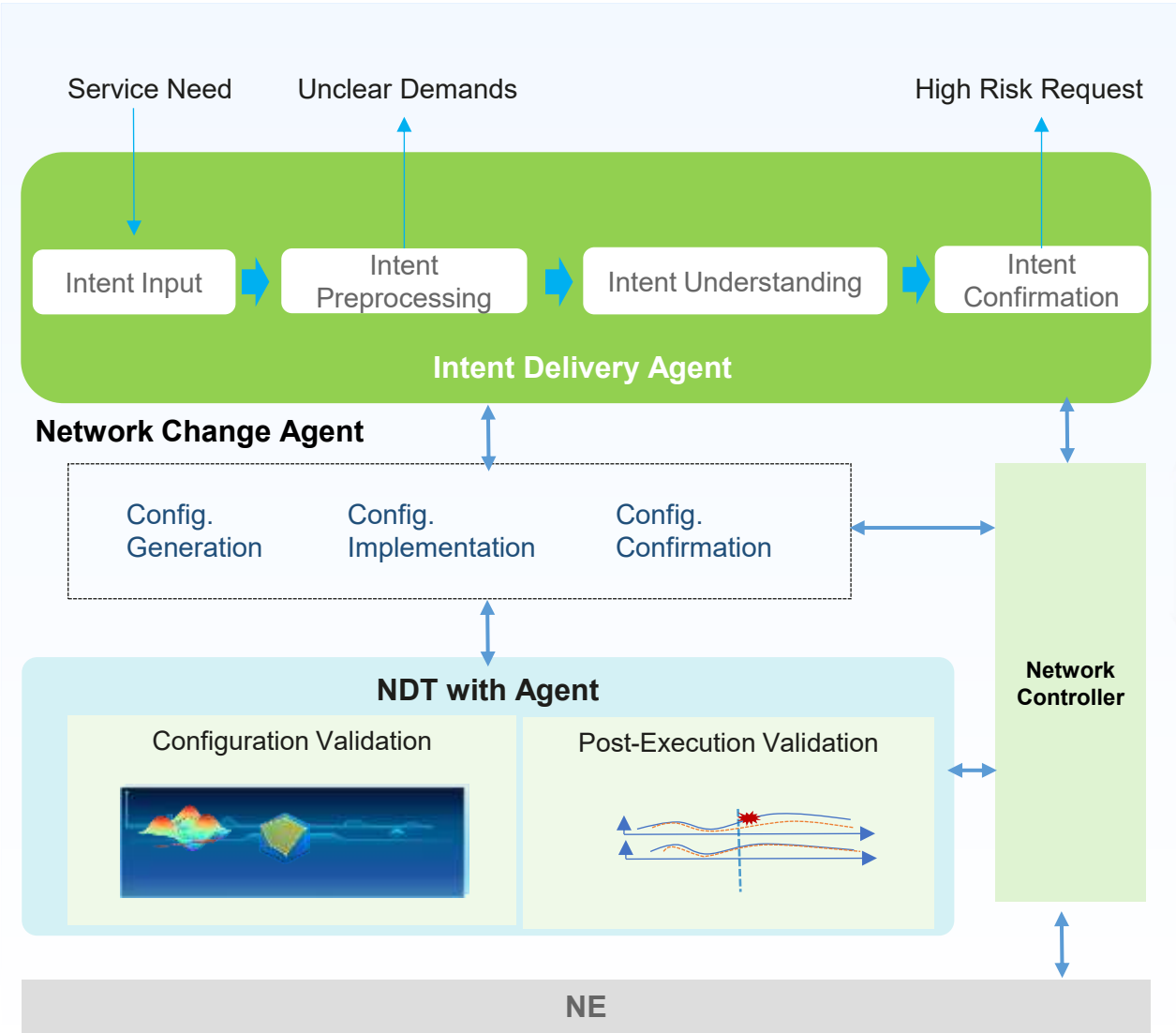


Using A2A-T protocol to deliver service intent, further research is required.

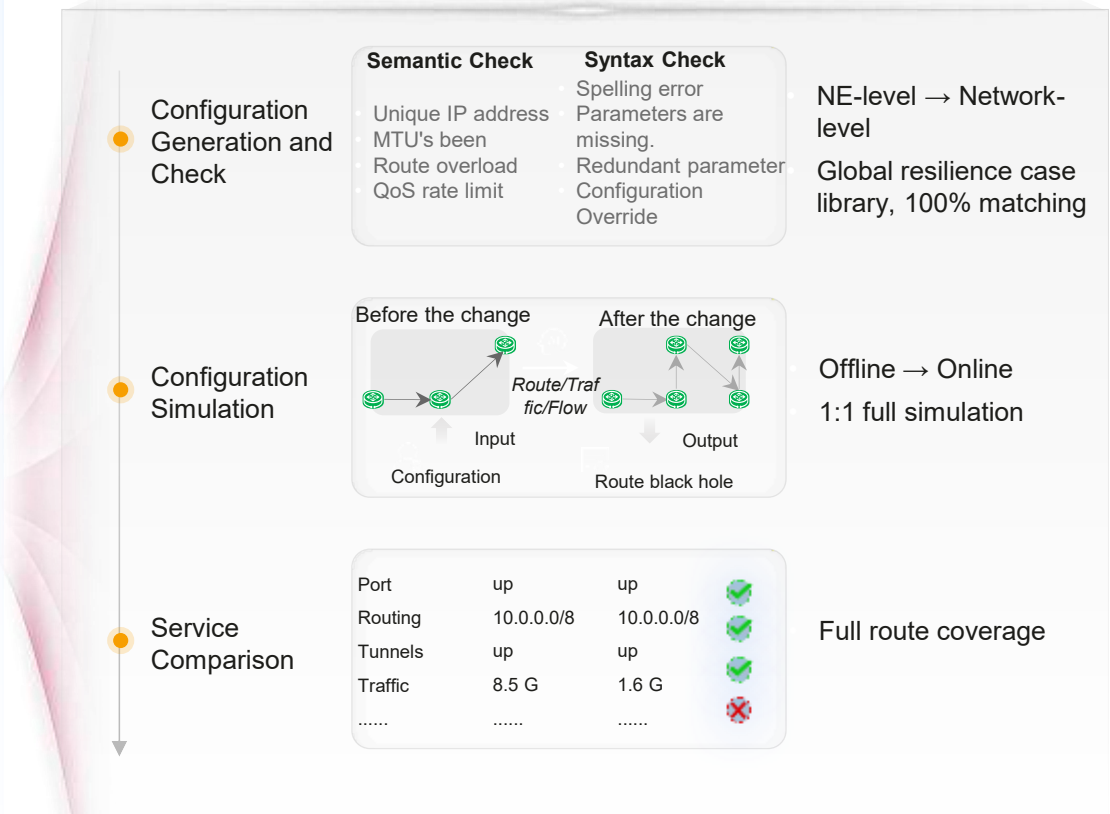


Integration of agent-related protocols (e.g., MCP) with traditional IP protocols.

Workflow and Values



Fully automated configuration generation, simulation and deployment.



Values

Critical Accidents: 0

Network Change Efficiency: rise by 90%

Future Work

Interface among agents

Transmit intention, trust and constraints.

- Extend A2A or MCP protocols. Adaptation for IP network.
- Drive NDT for simulation and verification.
- Align knowledge and semantic information.
- Observability.

Interface between DTN and Physical Network

Provide network data and information required by NDT agents.

- Intent-driven, bidirectional data transfer.
- Real-time data synchronization.
- Security.

Discussion